

Decision support for personalized therapy in implantable medical devices: A digital twin approach

Haochen Yang^a, Zhihao Jiang^{a,b,*}

^a School of Information Science and Technology, ShanghaiTech University, Shanghai, China

^b Shanghai Engineering Research Center of Intelligent Vision and Imaging, Shanghai, China

ARTICLE INFO

Keywords:

Decision support
Digital twin
Therapy customization
Reinforcement learning

ABSTRACT

Implantable Medical Devices (IMDs) offer timely therapeutic interventions for life-threatening conditions without disrupting patients' daily activities. Given the substantial variability in individual health conditions, customizing device therapies to suit specific patients and their evolving needs is crucial. Although current customization relies on modifying device parameter settings, no universally accepted clinical guidelines exist for this form of customization, particularly for patients with complex or rare conditions. To address this gap, we introduce a decision support framework that leverages expert system methodologies for personalized therapy in IMDs. This comprehensive expert system encompasses a dynamic digital twin of the patient, built using physiological data obtained from the IMD and additional wearable devices. The digital twin captures disease mechanisms and patient-specific physiological parameters, while a periodically trained Reinforcement Learning (RL) agent and specialized algorithms for the ongoing maintenance of the digital twin complete the expert system. Together, these elements facilitate periodic, informed therapy customization based on up-to-date patient conditions. Our approach includes novel algorithms for feature extraction, physiological modeling, and the maintenance of the digital twin. We substantiate the efficacy and safety of our digital-twin-based decision support framework through a case study focused on Implantable Cardioverter Defibrillators (ICDs). Experimental results on virtual patients confirm that our integrated framework surpasses traditional customization strategies in both therapeutic effectiveness and patient safety considerations.

1. Introduction

1.1. Implantable medical devices (IMDs)

Diseases such as cardiac arrhythmia and epileptic seizures manifest suddenly and can result in fatal consequences within minutes, making immediate treatment imperative (Peretto, Durante, Limite, Cianflone, et al., 2014). Implantable medical devices (IMDs), including Implantable Cardioverter Defibrillators (ICDs) and Deep-Brain Stimulators (DBS), are engineered to detect the onset of these diseases automatically and administer treatment without disrupting the daily activities of patients (Khan, Muntimadugu, Jaffe, & Domb, 2014). The accuracy of therapeutic decisions by the embedded software in these devices is vital for ensuring patient safety and well-being. Clinical studies on ICDs reveal that these devices exhibit a 98% sensitivity, suggesting that they address the majority of disease onsets, and a specificity exceeding 92%, indicating infrequent unnecessary treatments (Olde Nordkamp et al., 2015).

1.2. Need for personalized IMD therapy

However, due to the inherent variability among individual patients, some may face a significant number of misdiagnoses, which introduce substantial risks. For instance, for patients diagnosed with supra-ventricular tachycardia (SVT), the occurrence of inappropriate ICD therapy can reach up to 47% (Korte, Köditz, Niehaus, Paul, & Tebbenjohanns, 2004). Patients' disease conditions can also evolve over time. Devices employing a static therapeutic strategy do not adjust to the progression of the disease condition, thereby impacting therapeutic outcomes (Wang et al., 1991). There exists a pressing requirement for online therapy personalization in IMDs. This personalization should be: (1) tailored specifically to individual patients for optimal outcomes; and (2) capable of updates to accommodate evolving disease conditions.

At present, the personalization of therapies largely depends on the customization of device parameters during the implantation stage, a process heavily reliant on a physician's expertise.¹ Additionally,

* Corresponding author at: School of Information Science and Technology, ShanghaiTech University, Shanghai, China.

E-mail addresses: yanghch@shanghaitech.edu.cn (H. Yang), jiangzhz@shanghaitech.edu.cn (Z. Jiang).

¹ In the context of our research, customization refers to selecting from existing solutions, whereas personalization involves designing individualized solution.

adjustments to these parameters are typically made during periodic follow-up appointments, which may occur every few months. However, this schedule can be inadequate for addressing sudden changes in a patient's condition (Wang et al., 1991). While implantable medical devices and wearable devices can gather a wealth of personalized physiological data, it is impractical for physicians to sift through hours or days of such data to infer the patient's physiological condition. Several clinical trials (Moss et al., 2012; Olshansky et al., 2019; Wilkoff, Williamson, Holloman, et al., 2008) have been undertaken to discern optimal parameter configurations for various diseases. However, these findings are disease-specific, leaving a lack of comprehensive guidelines for parameter adjustments tailored to individual patients. Notably, incorrect parameter configurations have been implicated in patient fatalities (Wilkoff et al., 2008). There is a pressing requirement for an automated and thorough approach to customizing or personalizing IMD parameters. This process should be timely and effectively utilize the personal data collected from patients.

1.3. Challenges in automated IMD personalization

Safety Concerns: Ensuring patient safety is paramount, as IMDs regulate essential physiological functions and cannot rely on a trial-and-error approach. The automation of IMD parameter customization introduces periods where medical personnel are not directly involved in the decision-making process. This necessitates stringent safety measures within the system to mitigate risks.

Hardware Limitations: IMDs are constrained by their limited computational power and battery life, due to their embedded and minimally invasive design. This limitation challenges the direct implementation of complex, computationally intensive customization algorithms on these devices.

Data Constraints: The data storage capacity in IMDs is limited, restricting the volume of patient data that can be used for personalized algorithm development. Additionally, the lack of comprehensive, labeled data, resulting from limited patient observability, poses significant challenges in creating effective, patient-specific customization strategies.

1.4. Related work

1.4.1. Reinforcement learning in clinical applications

Identifying the optimal therapy strategies for specific patients can be abstracted as sequential decision problems, where Reinforcement Learning (RL) has shown significant promise. For instance, RL has been applied to support clinical decisions such as choosing optimal medication or dosage in treatments to potentially improve hospital survival rates (Lin, Stanley, Ghassemi, & Nemati, 2018; Wang, Zhang, He, & Zha, 2018), yet this approach, trained on large patient cohorts, might be overly conservative and not ideal for individual patient needs.

A notable case is Gao et al. (2020), where RL is directly integrated into IMD therapy personalization for Deep Brain Stimulation (DBS) in Parkinson's disease patients. However, the challenge lies in the computational demands of RL, which necessitates patients to carry additional hardware like laptops, contradicting the core principle of IMDs being unobtrusive.

1.4.2. Model-based reinforcement learning

The model-based Reinforcement Learning utilizes forward dynamics models to encapsulate generalized domain knowledge of the environment, enabling effective agent learning with limited data. Studies such as Barati, Kulkarni, Clark, and Barn (2022) and Wang et al. (2022) investigate the use of environment models as experience generators within RL frameworks, albeit outside the medical decision-making domain. While these studies used the term "digital twins", they primarily focus on generalized models without emphasizing customization, or they

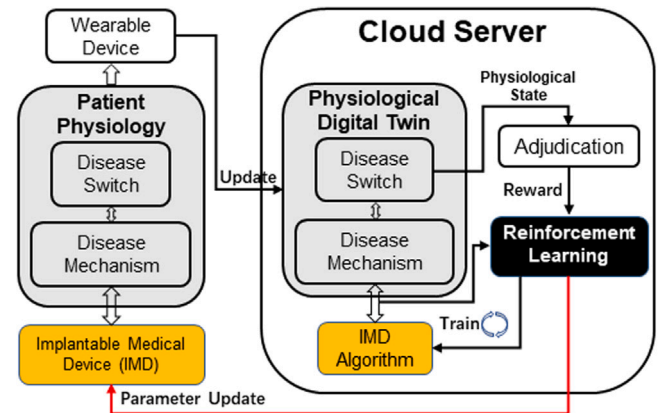


Fig. 1. Flowchart illustrating the periodic update cycle of the Digital-Twin-Based Decision Support Framework. The framework captures physiological data from IMDs and wearable devices, updates the patient's digital twin, and employs a Reinforcement Learning agent to optimize therapy parameters for IMDs.

overlook the necessity of continuous model updates. This limitation underscores a missed opportunity in fully exploiting the capabilities of digital twins for dynamic, real-time adaptation in contexts like IMD therapy customization.

1.5. Digital-twin-based IMD parameter customization

In this study, we present a digital-twin-based decision support framework for customizing parameters in implantable medical devices (IMDs). As depicted in Fig. 1, a patient's physiology is represented by (1) the physiological mechanism of the diseases, and (2) transitions between different disease states. IMDs, along with other wearable medical devices, capture physiological data from the patient. In recurring cycles (spanning days or weeks), the digital twin undergoes updates based on this data, allowing the reinforcement learning (RL) agent to train with the digital twin serving as its environment. Using insights from the refreshed patient state, the RL agent suggests an optimized parameter configuration for the IMD. Notably, our framework facilitates the RL agent's training on a virtual patient, mitigating data scarcity challenge. The timed-automata digital twin over-approximate the patient's behaviors, ensuring the RL agent's learned strategies are cautious and prioritizing safety. Since there is no pressing need for frequent parameter adjustments, executing the RL algorithm on a cloud server becomes feasible. This approach aims to deliver personalized treatments while ensuring patient safety.

In our prior paper (Lai, Yang, Gu, Chen, & Jiang, 2022), we introduced a preliminary framework for the customization of Implantable Cardioverter Defibrillator (ICD) parameters using RL, leveraging a mock digital twin of the patient's heart. While this earlier approach showed promise in identifying more effective parameter settings for various cardiac conditions, it operated under the assumption that a patient's digital twin was readily available. It did not delve into the complexities of how such a digital twin could be created or updated using personalized physiological data.

In this article, we broaden our customization framework to encompass a wide range of implantable medical devices (IMDs). Similar to ICDs, these IMDs are designed to invasively monitor physiological signals from patients, analyze these signals to extract features, and subsequently make therapeutic decisions. Complementing these devices, additional wearable technologies can gather more comprehensive physiological data, which assists in deducing the patient's health condition. We introduce a novel approach for constructing physiological models and a strategy for creating and maintaining patient-specific

Table 1
Signal sources of commercially-available IMDs.

	IMD signal sig^{inv}	Feature event σ^f	Therapy σ^T	Condition label σ^c	Wearable signal sig^{Ninv}
ICD	Electrogram	Depolarization	Defibrillation	VT/not VT	ECG (wave)
DBS (Epilepsy)	LFP/ECOG	Seizure	Pacing	Seizure/normal	IMU (tremor)
DBS (Parkinson)	LFP/ECOG	β power	Pacing	“On/Off”	IMU (gait)

digital twins. Utilizing these digital twins as a simulated environment, Reinforcement Learning (RL) agents are employed to determine the most effective parameter settings for the IMD tailored to individual patients. We once again utilize ICDs as a case study to demonstrate the effectiveness and safety of our proposed framework. Our experiments indicate that: (1) physiological attributes can be precisely derived from the relevant data; (2) even with sparse datasets, learning algorithms can adeptly construct and maintain a digital twin; (3) benchmarked against alternative methodologies, our approach consistently attains enhanced therapeutic results.

This paper’s main contributions include: (1) The introduction of a distinct digital-twin-centric framework for personalizing parameters in implantable medical devices (IMDs), offering decision support for physicians; (2) The proposal of an innovative loss function tailored to boost accuracy when extracting features from wearable devices; (3) The unveiling of a pioneering algorithm, anchored in genetic programming, aimed at both learning about and maintaining the digital twin; (4) An illustrative case study centered on Implantable Cardioverter Defibrillators (ICDs) to validate the merits of our framework.

2. Basics of implantable medical devices

2.1. IMD functionality

2.1.1. Feature extraction from invasive sensing

Implantable medical devices (IMDs) invasively capture physiological signals (sig^{inv}) from patients, which contains localized features (σ^f) that are crucial for diagnostic and therapeutic decisions. Let $\sigma_i^f \in \Sigma^f$ denotes the extracted feature event at time $i \in \mathbb{R}$. IMD software extracts *feature sequence* from physiological signals as:

$$w^f = IMD^{Feature}(sig^{inv}) \quad (1)$$

in which $w^f = (\sigma_1^f \sigma_2^f \dots \sigma_m^f)$. Owing to the constraints of embedded systems, IMD software employs straightforward rule-based algorithms for feature extraction.

2.1.2. Therapeutic decision making by IMDs

Therapeutic decision-making in implantable medical devices (IMDs) is typically based on predefined criteria or thresholds, focusing on identifying specific conditions or signals, rather than conducting complex quantitative analyses or numerical comparisons. Let $\Sigma^T = \{\text{Therapy, No Therapy}\}$ be the therapeutic decision options for each time period. The IMDs make therapeutic decisions based on the features identified from physiological signals:

$$w^T = IMD^{Decision}(w^f) \quad (2)$$

in which $\sigma_k^T \in \Sigma^T$ represents the therapeutic decision for the k th time period.

2.1.3. Adjudication of therapeutic decisions

The effectiveness of therapeutic interventions is contingent on the patient’s physiological condition, which can be expressed as:

$$w^c = (\sigma_1^c \sigma_2^c \dots \sigma_n^c)$$

in which $\sigma_j^c \in \Sigma^c$ represents the *condition label* for the j th time period. Therapeutic decisions are typically adjudicated offline by medical professionals:

$$\sigma^A = adj(\sigma^c, \sigma^T) \quad (3)$$

In this context, $\sigma^A \in \Sigma^A = TP, TN, FP, FN$ symbolizes true positive, true negative, false positive, and false negative labels for each therapeutic decision. Due to the critical nature of missed therapies in cases where they are needed, false negatives are unacceptable for IMDs. Consequently, there might be instances of false positives where therapies are delivered inappropriately.

2.2. Information from non-invasive sensing

Invasive signals from IMDs provide limited, localized health information and lack a holistic view of the patient’s health, making them insufficient for comprehensive therapy assessment.

On the other hand, wearable devices non-invasively gather data from the body’s surface, reflecting an aggregation of local physiological activities. This aggregated information offers a more comprehensive view of the patient’s physiological conditions, which helps determining whether the delivered therapies are appropriate. Wearable devices are typically designed for continuous, around-the-clock data collection. This constant data flow is particularly ideal for the creation and upkeep of digital twins.

Accurately extracting condition labels (w^c) and feature sequences (w^f) from non-invasive signals (sig^{Ninv}) is essential for creating a physiological digital twin. Traditional signal processing methods often fail to achieve the needed accuracy for such detailed and specific data extraction (Chen, Hua, Zhang, Liu, & Wen, 2020). Although machine learning algorithms provide the necessary precision, their high computational demands make them unsuitable for use in wearable devices for real-time therapeutic decisions.

2.3. Examples

Table 1 provides several examples to illustrate the operations of implantable medical devices (IMDs). One example is the implantable cardioverter defibrillator (ICD), which monitors intracardiac electrograms (EGMs) within the patient’s heart. It extracts features such as depolarization events in the atria and ventricles and delivers defibrillation in cases of ventricular tachycardia (VT). Additionally, external Electrocardiograms (ECGs) can be monitored by wearable devices, enabling the extraction of features and condition labels.

Another instance is the Deep Brain Stimulation (DBS) device, used for treating neurological diseases like seizures and Parkinson’s disease. DBS devices sense local field potential (LFP) or Cortical Electrocor-ticogram (ECOG) signals within the brain, extracting features like beta band power, and administer electrical pacing to specific brain regions to suppress abnormal activities. Concurrently, wearable devices equipped with Inertial Measurement Units (IMUs) can help determine the patient’s condition by detecting abnormal movements like tremor or freedzed gait.

3. Digital-twin-based parameter customization for implantable medical devices

3.1. Physiological modeling

The model used as digital twin should satisfy the following requirements:

Interaction: The model should be able to interact with device algorithms in closed-loop to serve as environment during reinforcement learning.

Adjudication: The model should capture disease mechanisms so that state of the model can be used determine whether a therapy or a missed therapy is appropriate.

Identifiability: Parameters of the model can be identified from physiological signals of specific patients with minimum ambiguities.

In the proposed framework, the patient's physiology is modeled by two distinct components:

3.1.1. Disease mechanism model (DMM)

It captures the disease mechanism needed for generating physiological signals or feature sequence w^f to IMDs and responding to device therapies. The DMM also reduces the amount of data needed for learning digital twins. Since the physiological mechanisms for different diseases varies, DMMs need to be specifically modeled for different IMDs.

3.1.2. Disease switch model (DSM)

The disease mechanism cannot and need not be fully identified from physiological data. The uncertainty associated with unidentified disease mechanism can be captured by stochastic models. In this paper, we propose the use of Markov Decision Process (MDP) as DSM, with states equal to the set of disease conditions:

$$S(DSM) = \Sigma^c$$

and the parameters of DSM consists of transition probability between physiological conditions:

$$\phi(DSM) = \{p(a|b) \mid \forall a, b \in S(DSM)\}$$

3.2. Digital twin learning algorithms

3.2.1. DSM learning

For a condition label sequence w^c of length T , the parameter of the disease switch model DSM can be estimated as:

$$\phi(DSM) = \{p(a|b) = \frac{\#(\sigma_{i-1}^c = b, \sigma_i^c = a)}{T-1}, \forall a, b \in S(DSM), 2 \leq i \leq T\} \quad (4)$$

3.2.2. DMM learning

For a feature sequence w^f of length T , the disease mechanism model DMM simulates an estimation of w^f :

$$DMM \models \widehat{w^f}$$

and the parameter of DMM can be estimated by minimizing the error of the simulated sequence:

$$\arg \min_{\phi(DMM)} L_{match}(\widehat{w^f}, w^f) \quad (5)$$

Since DMM may not be differentiable, genetic algorithm (Mohammadi, Asadi, Mohamed, Nelson, & Nahavandi, 2017) is used for parameter identification of DMM. Dynamic Time Warping (DTW) (Müller, 2007) is a commonly used loss function for similarity comparison of time series data. However, DTW focuses more on the morphology of the time series data and allows n-to-1 or 1-to-n matching of the events, which may not be appropriate for comparing sequences of timed events

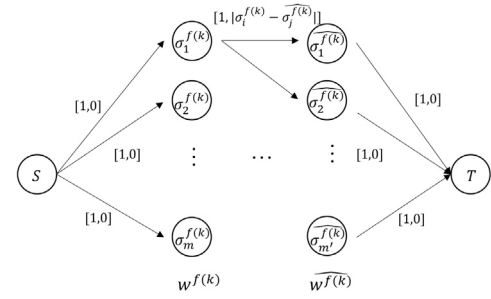


Fig. 2. The loss function for matching w^f and $\widehat{w^f}$ is defined as the minimum cost maximum flow of the bipartite graph, which ensures that the number of matches is no more than 1 and the time differences of matched events are minimized.

such as w^f . In this paper, we formulate the comparison problem as a minimum cost bipartite graph matching problem, and designed the loss function L_{match} . As show in Fig. 2, a bipartite graph is created for each $\sigma_i^{f(k)} \in \Sigma^f$. Each event from $w^{f(k)}$ represents a node on the left, and each event from model simulation $\widehat{w^{f(k)}}$ represents a node on the right. Each $\sigma_i^{f(k)}$ is connected to all nodes $\widehat{\sigma_j^{f(k)}}$ on the right, with capacity 1 and time difference as cost $|\sigma_i^{f(k)} - \widehat{\sigma_j^{f(k)}}|$. The similarity loss L_{match} of w^f and $\widehat{w^f}$ is defined as the minimum cost maximum flow of this bipartite graph, which ensures that the number of matches is no more than 1 and the time differences of matched events are minimized. The minimum cost maximum flow of graph $G = (V, E)$ can be calculated in $O(V^2 * \sqrt{E})$ using Highest Label Preflow Push (HLPP) algorithm (Cheriyian & Maheshwari, 1989).

3.3. Reinforcement learning for IMD customization

Reinforcement learning (RL) is widely used in recommendation tasks in e-commerce (Zhao et al., 2018). The environment is considered as a MDP which state s_t evolves according to actions a_t . The goal of the RL agent is to learn a policy $\pi(s)^*$ that maximizes the expected accumulated reward $r(s_t, \pi(s_t), s_{t+1})$ over the entire training horizon T . This is formally captured as

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{s_{t+1}|s_t, \pi(s_t)} \sum_{i=0}^T \gamma^i r(s_t, \pi(s_t), s_{t+1}) \quad (6)$$

in which γ is a discount factor indicating the relative weight of immediate and long-term rewards.

3.3.1. State

In the proposed framework, the state of the patient's physiology is formulated as:

$$s = \{w^f, w^c, w^T\}$$

in which w^f is the sensed feature sequence, w^c is the extracted condition label, and w^T is the sequence of therapy decisions.

3.3.2. Action

Making therapeutic decisions using reinforcement learning requires running RL algorithm on IMDs, which is computationally infeasible. In this paper, the action for RL algorithm is a parameter setting for IMD ($\phi(IMD) \in \Phi(IMD)$), which (1) fully utilizes the validated IMD software to ensure safety of the patient; and (2) reduces decision frequency which allows running RL algorithm on the cloud.

3.3.3. Reward

The goal of IMDs is to (1) provide therapies only when necessary; and (2) provide therapies in a timely manner. *Criteria* for necessary therapies correspond to the physiological condition of the patient. Therapies should be delivered when the criteria is met:

$$Cri(\sigma_i^c) == true$$

Therapy decisions should be *adjudicated* to determine whether the therapy is appropriate during the lifecycle of IMDs:

$$adj(\sigma_i^c, \sigma_i^T) = \begin{cases} TP & \sigma_i^T == T \ \& \ Cri(\sigma_i^c) == T \\ FP & \sigma_i^T == T \ \& \ Cri(\sigma_i^c) == F \\ TN & \sigma_i^T == F \ \& \ Cri(\sigma_i^c) == F \\ FN & \sigma_i^T == F \ \& \ Cri(\sigma_i^c) == T \end{cases} \quad (7)$$

In this paper, the reward for RL algorithm is set as:

$$r(s_t, \pi(s_t), s_{t+1}) = precision_t - penalty_t$$

in which the positive reward within a time period τ is the precision:

$$precision_t = \frac{\# \text{ of TP}}{\# \text{ of TP} + \# \text{ of FP}} \quad (8)$$

and the penalty within the same period is calculated based on the adjudication of the therapy decision sequence:

$$penalty_t = f(adj(w^T)) \quad (9)$$

3.3.4. RL algorithm

In this paper, we adopt the actor-critic (AC) architecture and the Proximal Policy Optimization (PPO) model (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017), which are popular in applications of reinforcement learning.

The digital twin of the patient's physiology is updated in each iteration t , which can be days or even weeks.

$$DMM_t = \mathcal{U}(DMM_{t-1}, w_t^f), DSM_t = \mathcal{U}(DSM_{t-1}, w_t^c)$$

The digital twin interacts with IMD software and generates simulated event sequences for large number of iterations:

$$DMM_t \models \widehat{w}^f, DSM_t \models \widehat{w}^c, \widehat{w}^T = IMD(\widehat{w}^f)$$

which are used as training data for RL algorithm:

$$\hat{s}_i = \{\widehat{w}_i^f, \widehat{w}_i^c, \widehat{w}_i^T\}, r(\hat{s}_i, \pi(\hat{s}_i), \hat{s}_{i+1})$$

Patient's data from the t th iteration is used to optimize a set of parameters for the next iteration:

$$s_t = \{w_t^f, w_t^c, w_t^T\}, a_{t+1} = \pi^*(s_t)$$

4. Parameter customization of ICDs

In this section, we use Implantable Cardioverter Defibrillator (ICDs) as case study to demonstrate the efficacy and safety of the proposed digital-twin-based therapy customization framework for IMDs.

4.1. ICD basics

Coordinated contractions of the atria and the ventricles of the heart are governed by the electrical conduction system of the heart. As shown in Fig. 3, a *Normal Sinus Rhythm* (NSR) starts from the self-triggered electrical depolarization in the SA node, triggering the contraction of the atria, triggering contraction of the ventricles after a prolonged delay in the AV node. Degeneration of heart tissue disrupt generation and conduction of electrical signals within the heart, which affects the heart rate and/or contraction synchrony. When the degeneration occurs above the ventricles, the disease is referred to as *Supra-Ventricular Tachycardia* (SVT), which is not a fatal disease but can affect the long-term health of the patient. On the other hand, patients with *Ventricular*

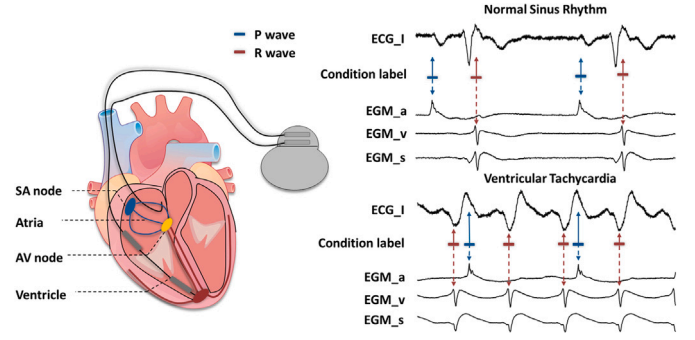


Fig. 3. Left: Lead configuration of a dual chamber ICD; Right: ECG and EGM of a patient during NSR and VT (AAEL (Ann Arbor Electrogram Libraries, 2010) file: SET1, A377a20.SIG).

Tachycardia (VT) have degeneration in the ventricles. The contraction rate and synchrony of the ventricles affected, which can be fatal, and the patient requires immediate treatment.

An ICD has 1–3 leads fixed onto the inside heart wall at different locations in the patient's heart. As shown in Fig. 3, a typical ICD has one lead in the right atrium and one lead in the right ventricle. Defibrillation shocks are delivered from two shock coils on the ventricle lead. Electrodes on these leads can sense local electrical activities, which are referred to as Electrogram (EGM) signals. The ICD extract contraction events of the atria (Ain) and the ventricles (Vin) from the EGMs:

$$\Sigma^f = \{Ain, Vin, -\}$$

which can be used to diagnose the patient's conditions for the atria and the ventricles:

$$\Sigma^{c(A)} = \{NSR, AT, AF\}, \Sigma^{c(V)} = \{NSR, VT, VF\}$$

in which AT , VT and AF , VF represent *tachycardia* and *fibrillation* in the atria and the ventricles. The patient's heart condition is the combination of chamber conditions, and the goal is to ensure defibrillation therapies are delivered when the patient is in fatal VT and withhold therapies otherwise.

$$\Sigma^c = \Sigma^{c(A)} \times \Sigma^{c(V)}, \Sigma^T = \{T, F\}$$

Device manufacturers have developed complex algorithms to identify fatal VTs. Although the algorithms from different manufacturers are slightly different due to IP reasons, features used in common include but not limited to: (1) intervals between ventricular events; (2) duration of a heart condition; (3) synchronization between atria and the ventricles; and (4) morphological features of EGMs. The performance of ICDs is evaluated using *sensitivity* (Se) and *specificity* (Sp) defined as:

$$Se = \frac{\#TP}{\#TP + \#FN}, Sp = \frac{\#TN}{\#TN + \#FP}$$

In this paper, we use the VT/SVT discrimination algorithm from Medtronic, which has shown good sensitivity and specificity in clinical trials (Gold et al., 2012). The specification of the algorithm was extracted from clinical literature and was implemented by Jiang et al. in Jiang et al. (2016). Modern ICDs can achieve 98%+ sensitivity and 92%+ specificity in clinical studies with large groups of patients (Olde Nordkamp et al., 2015). However, for specific patient the specificity can be as low as 53%, with a large portion of therapies inappropriate (Korte et al., 2004). There is need for personalized therapy strategy to improve ICD performance on specific patients.

4.2. Physiological signal processing

Ain and Vin events can be easily extracted from EGM signals. However, as an embedded system, IMDs cannot save EGM signals for

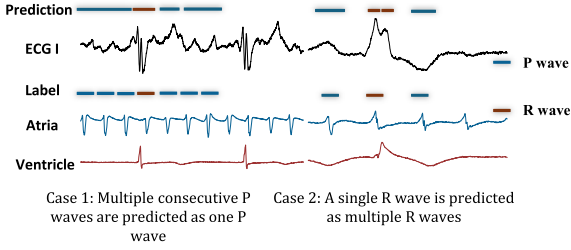


Fig. 4. Two examples in which frame-wise accuracy does not imply wave-wise accuracy (AAEL (Ann Arbor Electrogram Libraries, 2010) file: SET1, A182430.SIG).

extended period of time. Additional signal sources are needed in order to obtain adequate data for digital twin training. *Electrocardiogram (ECG)* is a commonly used technique for diagnosing heart conditions. As shown in Fig. 3, electrical activities at different locations of the heart are cumulatively reflected on Lead II ECG. The P wave and R wave capture the strong electrical activities caused by the contractions of the atria and the ventricles, which correspond to *Ain* and *Vin* events in EGMs. Currently, ECG can be measured using wearable ECG patches, which can save ECG signals 24/7 for extended period of time. In the paper, we extract condition label sequence w^c and feature sequence w^f from ECGs, which are later used for training digital twins.

4.2.1. Condition labeling w^c

There are 3 atrial conditions and 3 ventricular labels, which correspond to 9 combined labels in Σ^c as heart conditions. Therefore heart condition labeling can be formulated as two 3-class classification problems. In this paper, we use deep learning method Long Short-term Memory (LSTM), which has been successful in heart rhythm classification using ECG (Chen et al., 2020).

4.2.2. Feature extraction w^f

Researchers have studied ECG segmentation using traditional signal processing techniques (Rahul, Sora, & Sharma, 2021; Yochum, Renaud, & Jacquir, 2016) and deep learning (Abrishami, Han, Zhou, Campbell, & Czosek, 2018; Moskalenko, Zolotykh, & Osipov, 2019). Each data point (frame) in a single channel ECG can be classified into 3 categories in $\Sigma^f = \{Ain, Vin, -\}$. Frame-wise cross entropy was traditionally used as loss function to evaluate the performance of segmentation:

$$L_{CE}(w^f, w_{GT}^f) = \sum_{i=1}^N \sum_{j \in \Sigma^f} -(\sigma_{GT(i)}^f == j) * \log(p(\sigma_i^f == j)) \quad (10)$$

However, the pattern and timing of features are important for IMD algorithms. As shown in Fig. 4, mixing two waves into one or identifying one wave as two can significantly affect the perceived heart rate. In this paper, consecutive frames with the same classification are considered as a wave. It is important to introduce wave-wise accuracy which takes into account the number of waves as well as pattern of the waves.

Connectionist Temporal Classification (CTC) (Graves, Fernández, Gomez, & Schmidhuber, 2006) is a loss function used for labeling unsegmented sequential data, and has been successfully used in speech recognition (Kim, Hori, & Watanabe, 2017). For successive tokens in w_{GT}^f such as:

$$w_{GT}^f = \{-, Ain, Ain, -, Vin, Vin, -, Ain, -\}$$

The corresponding pattern is defined as:

$$k(w_{GT}^f) = \{Ain, Vin, Ain\}$$

The CTC loss is defined as:

$$L_{CTC}(w^f, w_{GT}^f) = \sum_{k(C)=k(w_{GT}^f)} p(C|w^f) = \sum_{k(C)=k(w_{GT}^f)} \prod_{i=1}^N p(\sigma_i^f == c_i), \quad c_i \in \Sigma^f \quad (11)$$

In this paper, we proposed a CTC-augmented loss function as follows:

$$L_{match} = \begin{cases} 0.9 * L_{CE} - 0.1 * L_{CTC} & F1_{P_frame} < 0.8 \\ 0.5 * L_{CE} - 0.5 * L_{CTC} & 0.8 \leq F1_{P_frame} < 0.9 \\ 0.1 * L_{CE} - 0.9 * L_{CTC} & F1_{P_frame} \geq 0.9 \end{cases} \quad (12)$$

which relies on the cross entropy loss L_{CE} initially to achieve high frame-wise accuracy, and gradually increase the weight of the CTC loss L_{CTC} to achieve high wave-wise accuracy.

4.3. Electrophysiological (EP) model of the heart

In this paper, we propose a electrophysiological heart model as the model structure of the digital twin. As shown in Fig. 5, the electrophysiological heart model consists of a disease mechanism model (DMM), and a stochastic disease switch model (DSM):

4.3.1. The disease switch model (DSM)

As shown in Fig. 5, DSM consists of two Markov Processes $\mathcal{M}_A$ and $\mathcal{M}_V$ for the atria and the ventricles with state space:

$$S(\mathcal{M}_A) = \{NSR, SVT, AF\}, S(\mathcal{M}_V) = \{NSR, VT, VF\}$$

When a therapy from ICD occurs, the disease switching model switch to (NSR, NSR) state. The parameters of the DSM is the transition probability among states.

4.3.2. The disease mechanism model (DMM)

In Jiang, Pajic, and Mangharam (2012), Jiang et al. proposed an electrophysiological model of the heart, which has been validated by physicians. Self-depolarization and refractory properties of heart tissue can be modeled using node automata, and conduction delays between heart issues can be modeled using path automata. In this paper, we adopt the heart model framework and model the heart using 3 nodes and 2 paths (Fig. 5):

$$DMM = \{N_1, P_1, N_2, P_2, N_3\}$$

The atrial (N1) and ventricular nodes (N3) generate atrial and ventricular events *Ain* and *Vin* as input to the ICD. The ventricular node (N3) can distinguish whether a ventricular event is triggered by healthy A-V conduction or unhealthy self-depolarization/reentry. The AV node (N2) and the two paths (P1, P2) model the refractory behaviors of the AV node and the conduction delays between the atria and the ventricles. Each node automaton has 4 parameters:

$$\phi(N_i, c) = \{RestMin^c, RestMax^c, ERPMin^c, ERPMac^c \mid c \in S(\mathcal{M}_A) \times S(\mathcal{M}_V), 1 \leq i \leq 3\}$$

Each path automata has 2 parameters:

$$\phi(P_j, c) = \{CondMin^c, CondMax^c \mid c \in S(\mathcal{M}_A) \times S(\mathcal{M}_V), 1 \leq j \leq 2\}$$

For each combined atrial and ventricular state in the DSM, there is a set of parameters in the DMM:

$$\phi(DMM^c) = \{\phi(N_i, c), \phi(P_j, c) \mid \forall 1 \leq i \leq 3, 1 \leq j \leq 2\}$$

Upon state changes in the DSM, the parameters in the DMM are adjusted accordingly.

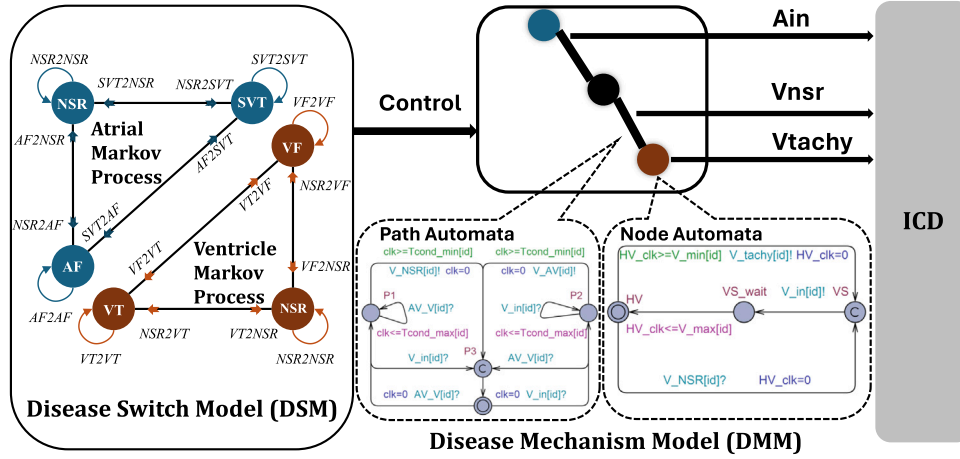


Fig. 5. Electrophysiological (EP) model of the heart.

4.4. Reinforcement learning

4.4.1. Reward

We define d_t as the length of consecutive false-negatives (FN), which indicates the time delay for an ICD therapy since a VT has occurred:

$$\text{penalty}_t = \begin{cases} 0 & d_t \leq 20 \text{ s} \\ 0.3 & 20 \text{ s} < d_t \leq 30 \text{ s} \\ 0.4 & 30 \text{ s} < d_t \leq 40 \text{ s} \\ 0.6 & 40 \text{ s} < d_t \leq 60 \text{ s} \\ 1 & 60 \text{ s} < d_t. \end{cases} \quad (13)$$

4.4.2. Action

Individual patients are different in *type*, *rate*, and *occurrence frequency* of arrhythmia. Physicians can adjust parameters in ICDs for personalized therapy strategy based on prior knowledge of the patient's condition. In this paper, the following parameters are considered during parameter customization:

- VT_thresh ([280, 650] ms): A ventricular event is considered as a VT event if the interval to the previous ventricular event is less than VT_thresh .
- VF_thresh ([240, min(400, VT_thresh)] ms): A ventricular event is considered as a VF event if the interval to the previous ventricular event is less than VF_thresh .
- SVT_limit ([240, 650] ms): If the median of intervals between ventricular beats in an episode (24 beats) is less than SVT_limit , the episode is double-checked for SVTs.

Inappropriate parameter settings may cause false-positives or false-negatives. For instance, if VT_thresh is set lower than the cycle length of VTs, the VTs will not be detected and cause false-negatives. If SVT_limit is set above the cycle length of SVTs, the SVTs will be treated as VTs and cause false-positives.

4.4.3. Training

In our current configuration, the length of each state (one iteration) is $\tau = 1$ week. The RL algorithm trains on the digital twin for $T = 1000$ weeks virtually (2.78 hours in real time).

4.4.4. System operation

The first three weeks serve as the “cold start” period, during which ECG signals from the patient are collected, and a digital twin is learned. At the end of Week i ($i > 3$), the digital twin is updated weekly using patient data from Week $[1, i]$, with $1/\text{week}$ increasing weight on the more recent data. The RL algorithm trains on the digital twin of the

patient and suggest a new set of ICD parameters using patient data from Week i , which is programmed into the ICD at the beginning of Week $i + 1$.

5. Experiments

5.1. ECG databases

There are publicly available ECG databases with labels such as QTdataset and LUDB dataset (Kalyakulina et al., 2018; Laguna, Mark, Goldberg, & Moody, 1997). However, these databases contains mostly normal heart conditions which do not reflect complex heart conditions the ICD is designed to perform in. The AAEL database (Ann Arbor Electrogram Libraries, 2010) was created to assist developing and validating ICD algorithms. EGMs were collected from participating patients to mimic the input signals to the ICDs, along with two channels of ECGs. Unlike publicly available ECG databases in which tachycardia need to spontaneously occur, atrial and/or ventricular tachycardia were manually induced in AAEL to provide more complete information of a patient. Therefore in this paper, we use AAEL as dataset for validating the proposed framework.

5.2. Virtual patients

Due to the availability of long (multiple days) ECG signals for specific patients, especially the ones with evolving heart conditions, it is currently infeasible to validate our approach on real patient data. In this paper, we use virtual patients to validate our cardiac digital twin framework. Each virtual patient is modeled by a disease mechanism model (DMM) and a disease switch model (DSM), which generates $s_t = \{w_t^f, w_t^c, w_t^T\}$. We first generate two base DSMs and one base DMM using data from clinical literature. Random virtual patients can then be generated by sampling model parameters “within” or “around” the parameters of these base models.

5.2.1. Base DSM

Based on meta data collected from two clinical trials for ICDs, we created two DSM models which serve as base DSMs for our validation experiments. During the 3-month follow-up of the MADIT-RIT trial (Moss et al., 2012), the ICD had a total of 517 appropriate treatments in approximately 113 patients, and 998 inappropriate treatments in approximately 135 patients in the Medtronic Trial (Wilkoff et al., 2001). Since the inappropriate therapy rate for ICDs is approximately 10%–20% (Olde Nordkamp et al., 2015), the true number of SVT/AF occurrences can be estimated using five times the number of inappropriate treatments. Since the sensitivity of ICD is close to 100%, the number of appropriate treatments can be used as the number of occurrences of VT/VF. The average frequency of disease onset for the two virtual patients are shown in Table 2.

Table 2

The average frequency of disease onset for the two virtual patients from two clinical trials.

ICD Trials	VT/VF	AT/AF
MADIT-RIT [23]	1.5/month	2.3/month
Medtronic Trial [35]	4.3/month	3.5/month

Table 3

The comparison of feature extraction algorithms on the QT dataset & LUDB.

Methods		QT Database&LUDB			
		P frame accuracy	R frame accuracy	P wave accuracy	R wave accuracy
Bi-LSTM [1]	Sensitivity(%)	0.91	0.97	0.97	0.99
	PPV(%)	0.93	0.96	0.61	0.83
	F1(%)	0.92	0.97	0.75	0.90
	Diff	-	-	4.9	1.9
Unet [22]	Sensitivity(%)	0.93	0.97	0.92	0.96
	PPV(%)	0.92	0.93	0.69	0.81
	F1(%)	0.92	0.95	0.78	0.88
	Diff	-	-	4.3	2.0
CTC- Aug	Sensitivity(%)	0.92	0.97	0.92	0.99
	PPV(%)	0.96	0.97	0.91	0.93
	F1(%)	0.94	0.97	0.92	0.96
	Diff	-	-	0.20	0.60

Table 4

The comparison of ECG segmentation algorithms on the AAEL dataset.

Methods		AAELPR Database			
		P frame accuracy	R frame accuracy	P wave accuracy	R wave accuracy
Bi-LSTM [1]	Sensitivity(%)	0.90	0.95	0.58	0.69
	PPV(%)	0.86	0.98	0.41	0.80
	F1(%)	0.87	0.96	0.48	0.74
	Diff	-	-	9.6	8.3
Unet [22]	Sensitivity(%)	0.91	0.97	0.42	0.73
	PPV(%)	0.88	0.94	0.31	0.54
	F1(%)	0.89	0.95	0.35	0.62
	Diff	-	-	11.2	9.9
CTC- Aug	Sensitivity(%)	0.88	0.96	0.85	0.92
	PPV(%)	0.86	0.93	0.80	0.93
	F1(%)	0.87	0.95	0.83	0.93
	Diff	-	-	2.3	1.6

5.2.2. Base DMM

A base DMM of the heart is generated from all EGM data from AAEL database, using algorithm discussed in Section 3.2.2. The resulting ranges (min and max) of DMM parameters are large, as the EGM data are from different patients. The parameter ranges of a random virtual patient's DMM are subsets of the parameter ranges of the base DMM.

5.3. Physiological signal processing

5.3.1. Feature extraction

LSTM (Abrishami et al., 2018) and UNet (Moskalenko et al., 2019) are two deep learning frameworks widely used for ECG segmentation. As shown in Table 3, both frameworks using the traditional cross entropy loss function perform well on QTdataset and LUDB dataset (Kalyakulina et al., 2018; Laguna et al., 1997) with mostly NSR conditions. However on AAEL dataset (Table 4), the two frameworks perform reasonably well in frame-wise evaluation, but not in wave-wise evaluation. Fig. 4 shows two cases in which the frame-wise accuracy is high and the wave-wise accuracy is low. In order to correctly identify P and R waves, the patterns of P and R waves should be taken into account. As shown in Table 4, our proposed algorithm using Bi-LSTM framework and CTC-augmented loss function achieved superior wave-wise accuracy while maintaining high frame-wise accuracy compared to traditional algorithms.

5.3.2. Condition labeling

As shown in Table 5, LSTM achieved 98%+ sensitivity and specificity in ventricular conditions, and 95%+ in challenging atrial conditions.

Table 5

Condition labeling result of the AAEL database.

Rhythm	Atrial Rhythm Accuracy			Ventricular Rhythm Accuracy		
	Se(%)	PPV(%)	F1(%)	Se(%)	PPV(%)	F1(%)
(NSR, NSR)	1	1	1	1	1	1
(NSR, VT)	1	1	1	1	1	1
(NSR, VF)	0.98	1	0.99	1	1	1
(AT, NSR)	1	1	1	1	1	1
(AF, NSR)	0.99	1	0.99	1	1	1
(AF, VT)	0.96	1	0.98	0.98	1	0.99
(AF, VF)	0.95	1	0.97	0.98	0.98	0.99

5.4. Validation of the cardiac digital twin

5.4.1. Learning of the disease switch model (DSM)

A learned DSM from the ECG can be represented as $\hat{\mathcal{M}}_A$ and $\hat{\mathcal{M}}_V$ with probability transfer matrix $\phi(\hat{\mathcal{M}}_A)$ and $\phi(\hat{\mathcal{M}}_V)$. The difference between $\mathcal{M}$ and $\hat{\mathcal{M}}$ can be quantified as the cross entropy loss between two probability matrix. Fig. 6a shows the cross entropy loss for both virtual patients over 100 days. The results indicate that the DSM model can be learned with small difference from the ground truth with around 1 month of patient data.

5.4.2. Learning of the disease mechanism model (DMM)

The DMM is modeled in timed automata (Alur & Dill, 1994). Existing methods on identification of timed automata identify both the structure and the parameter (Tappler, Aichernig, Larsen, & Lorber, 2019; Verwer, 2010), which is not suitable in our application with known model structure. Parameter identification of timed automata with known model structure is difficult since timed automata are not differentiable. Some gradient-free optimization methods can be applied to this problem. In this project, we used genetic algorithms to identify the parameters of the DMM. The genetic algorithm (Mohammadi et al., 2017) was used with population size 1000, 400 iterations and the mutation probability of 0.1.

A ground truth DMM DMM_g^c with parameters $\phi(DMM_g^c)$ generates contraction event sequence w_c^f during heart condition c , and a digital twin DMM_i^c is learned with parameters $\phi(DMM_i^c)$ at Day i . Instead of directly comparing parameters $\phi(DMM_g^c)$ and $\phi(DMM_i^c)$, we validate the learning algorithms by checking whether the learned DMM DMM_i^c and the ground truth DMM DMM_g^c can be distinguished from the ICD's perspective. Short (10 s) feature sequences w^f and $\hat{w}^f$ can be obtained by simulating DMM_g^c and DMM_i^c , which are then examined by the ICD algorithm. If the decision x_g and x_i agrees,² DMM_g^c and DMM_i^c are considered as indistinguishable.

Since there are non-determinism in DMM_g^c and DMM_i^c , agreement in a single simulation can occur by chance. Sequential Hypothesis Testing (David et al., 2011) was adopted in order to obtain a statistical significant result. The simulation above is performed multiple times until statistical significance level has been reached. The null hypothesis H_0 and the alternative hypothesis H_1 can be formulated as:

$$H_0 : \mathbb{P}(DMM_g^c \text{ and } DMM_i^c \text{ are indistinguishable}) < \gamma$$

$$H_1 : \mathbb{P}(DMM_g^c \text{ and } DMM_i^c \text{ are indistinguishable}) \geq \gamma$$

in which $\gamma = 0.95$. The number of simulations increases until H_1 is either accepted or rejected. The statistical parameters in the experiment are: significance level $\alpha = 0.05$, Type II error rate $\beta = 0.05$, indifference probability $\gamma = 0.95$, and probability bound $\mu = 0.01$.

For each heart condition $c \in S(\mathcal{M}_A) * S(\mathcal{M}_V)$, 100 DMM_g^c models were sampled and learned separately. Fig. 6b illustrate the percentage

² Whether there exists at least one therapy in the duration of w^f

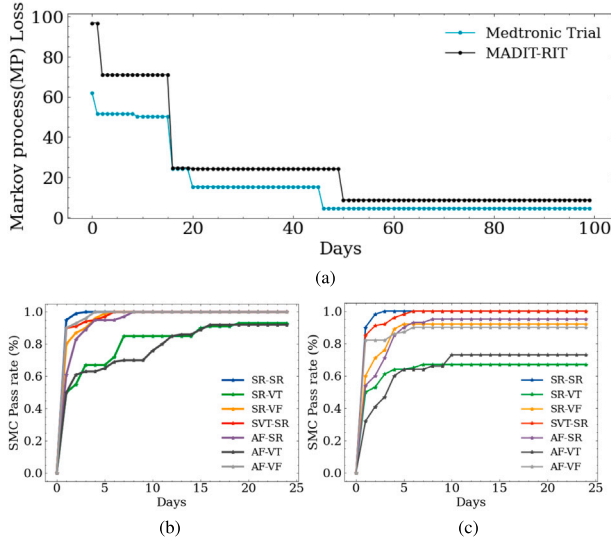


Fig. 6. (a) The DSM can be learned with high accuracy within 100 days; (b) and (c) The matching-based loss performs better than the DTW-based loss in DMM learning.

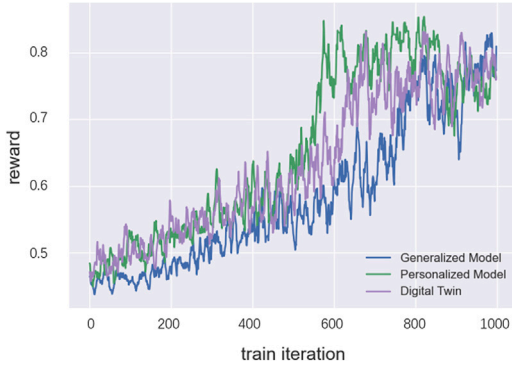


Fig. 7. Learning curves for model-based RL training.

of DMM_i^c that are indistinguishable from DMM_g^c at Day i for each heart condition c , with bipartite graph matching as loss function in the genetic programming learning algorithm. Fig. 6c illustrate the result using DTW as loss function. The results show that (1) the learned digital twin became increasingly indistinguishable from the ground truth with more data, and stabled after 10 days; and (2) the bipartite graph matching loss performs better than traditional DTW loss.

5.5. Validation of parameter customization

In this section, we validate the efficacy and safety of the digital-twin-based parameter personalization framework. 100 virtual patients with different DSM and DMM parameters were sampled based on the base DSM and DMM discussed in Section 5.2. Each virtual patient interacted with ICD algorithm for 600 weeks (11 years) under 4 different ICD parameter schemes:

1. **Baseline (BL) Scheme:** Off-the-shelf default ICD parameter setting is used throughout the experiment as baseline.
2. **Generalized Model (GM) Scheme:** A general model with base DSM and base DMM introduced in Section 5.2 was used as environment in RL algorithm training for 1000 weeks. ICD parameters for the r th iteration were updated as $\pi_{GM}^*(\{w_i^f, w_i^c, w_i^T\})$.
3. **Personalized Model (PM) Scheme:** A personalized model consists of DSM and DMM was created using patient data from

the first 3 weeks. The model was then used as environment to train RL algorithm for 1000 weeks. ICD parameters for the r th iteration were updated as $\pi_{PM}^*(\{w_i^f, w_i^c, w_i^T\})$.

4. **Digital Twin (DT) Scheme:** The personalized model was updated weekly after week 3 as DT_i , which was then used to train RL algorithm for 1000 weeks every iteration. ICD parameters for the r th iteration were updated as $\pi_{DT_i}^*(\{w_i^f, w_i^c, w_i^T\})$.

The learning curves for the model-based parameter scheme is shown in Fig. 7. The experimental results are shown in Fig. 8. Fig. 8(a) shows the cumulative reward (precision-penalty) for each patient over 600 iterations. Figs. 8(b) and 8(c) illustrate the total reward and penalty of a specific virtual patient over 600 iterations (weeks). The following statements can be interpreted from the results:

- **Model-based RL training is desired:** RL algorithm requires large amount of data to achieve reasonable reward, which is infeasible in real life. For instance, as shown in Fig. 7, 1000 weeks (19 years) of data from the patient is needed to achieve 0.8 reward, which equivalents to around 90% precision and therapies are delivered 20 s after disease onset.
- **Dynamic parameter setting is desired:** The baseline scheme with default ICD parameter setting has large variation in terms of cumulative reward, which reflects the large variability among patients. The generalized model scheme dynamically adjusts ICD parameters, which improve the lower bound of cumulative reward. However, the generalized model scheme is still “conservative” in the sense that extreme heart conditions are considered during parameter setting even if the specific patient does not exhibit those conditions.
- **Personalized parameter setting is desired:** The personalized model scheme demonstrated significant improvement over the generalized model scheme by 37.7% ($p < 0.01$). The PM scheme also demonstrated less “oscillation” in the overall reward (Fig. 8(b)), and less missed therapies (Penalty==1 in Fig. 8(c)).
- **Digital Twin of the patient is desired:** Although the digital twin scheme only have 6.5% improvement over the personalized model scheme ($p < 0.01$), with updated digital twin, there is no missed therapies after 150 weeks (Penalty==1 in Fig. 8(c)), which can significantly improve patient safety.

5.6. Performance on evolving disease

The patient’s physiological model was assumed to be unchanged in the previous experiment. However, disease parameters may change and new disease condition may develop, which should be adapted to by the parameter scheme.

The physiological conditions of the patients change at the 8th week during the experiment, which should reflect quickly in the digital twin and the corresponding ICD parameters. The condition changes are modeled by changing $\phi(DMM_g^c)$ in the virtual patient. We evaluate the influence of arrhythmia occurrence rate by comparing results using two DSM_g referenced from the MADIT-RIT and Medtronic trials introduced in Section 5.2.

5.6.1. Fast VT to slow VT

As shown in Fig. 9(a), the VT cycle length parameter of the virtual patient was [320,350] ms in the beginning, which was changed to [365,395] ms at Week 8 (region bounded by dashed black lines). After the 3-week cold start period, the digital twin updated weekly, which is illustrated by the VT cycle length parameter of the digital twin (blue bars). For VT, the most important ICD parameter is the VT_{thresh} , and VT cycle lengths below the threshold are considered as VT (black dash as parameter value, and green region indicates coverage). As shown in Fig. 9(a), the digital twin parameters quickly adapted to the ground

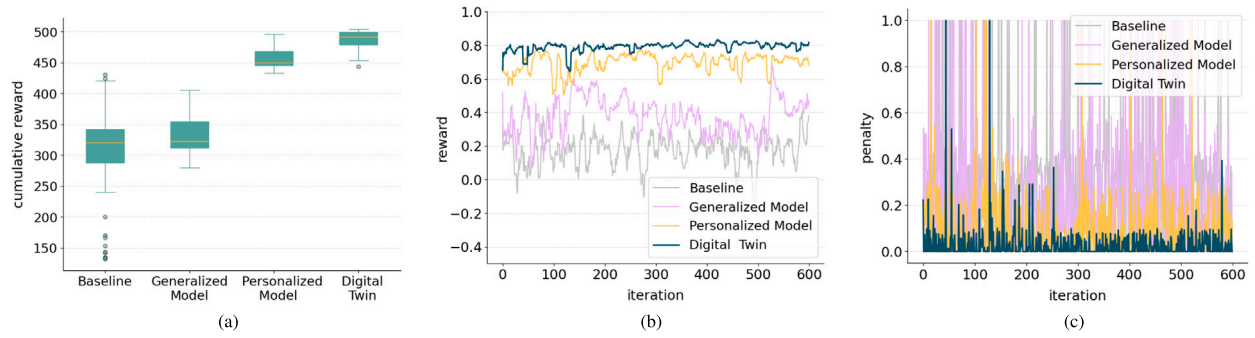


Fig. 8. (a) cumulative reward over 600 iterations for 100 patients under different parameter schemes; (b)(c), reward and penalty for a specific patient under different parameter schemes.

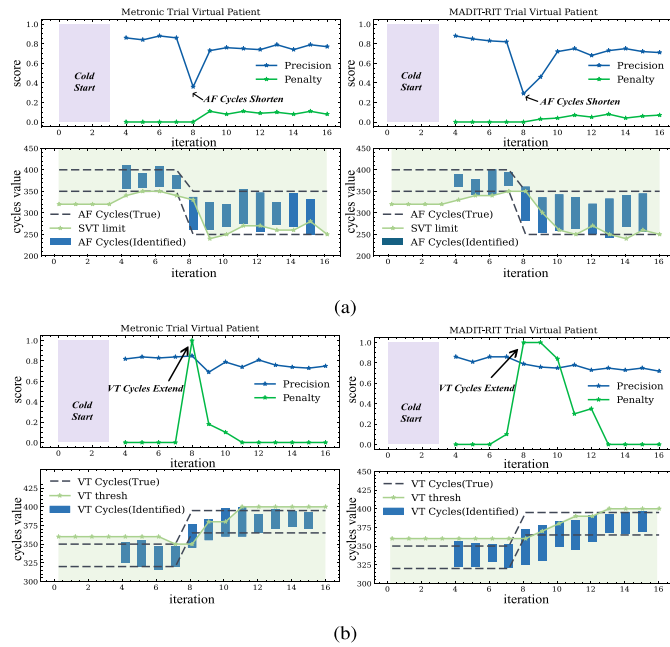


Fig. 9. (a) Digital twin and learned ICD parameters quickly adapted to newly developed slow VT; (b) for newly developed fast SVTs, reducing penalties was prioritized by sacrificing precision.

truth within one or two weeks, and the RL algorithm can always find the VT_thresh parameter that covers the VT cycle length parameter.³

5.6.2. Slow SVTs to fast SVTs

As shown in Fig. 9(b), the AF cycle length parameter of the virtual patient was [350,400] ms in the beginning, which was changed to [250,350] ms at Week 8. For SVT, the most important ICD parameter is the SVT_limit , and ventricular cycle lengths above the threshold are considered by the SVT/VT discrimination algorithm. Similar to the previous case, the digital twin parameter quickly adapt to the ground truth within one or two weeks, and the RL algorithm can always find the SVT_limit parameter that covers the AF cycle length parameter of the digital twin. Note that the penalty increased after the cycle length change, indicating there are delayed therapies during VT. In order to prioritize VT treatment, the RL algorithm decided not to cover some AF cycle length.

6. Summary and future work

In this paper, we propose a digital-twin-based therapy customization framework for IMDs. We have demonstrated with experiments that proposed framework can provide better therapeutic results and quickly adapt to disease development. In the ICD case study, an improved ECG segmentation algorithm was proposed, which has demonstrated better performance. For patients underwent EP testing procedures, additional information of the patients' heart can be obtained from EGM signals, which can be used to create more detailed cardiac digital twin of the heart. Methods proposed in Wu, Tang, Kang, and Jiang (2021) can potentially be introduced as future work, which can further optimize ICD parameters.

CRedit authorship contribution statement

Haochen Yang: Methodology, Software, Validation, Formal analysis, Investigation, Data curation. **Zhihao Jiang:** Conceptualization, Resources, Writing – review & editing, Supervision, Project Administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was supported by National Natural Science Foundation of China Young Scholar Program 62103279.

Statement: During the preparation of this work the author(s) used ChatGPT in order to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

- Abrishami, H., Han, C., Zhou, X., Campbell, M., & Czoszek, R. (2018). Supervised ECG interval segmentation using LSTM neural network. In *Proceedings of the international conference on bioinformatics & computational biology (BIOCOMP)* (pp. 71–77). The Steering Committee of The World Congress in Computer Science, Computer ...
- Alur, R., & Dill, D. L. (1994). A theory of timed automata. *Theoretical Computer Science*, 126, 183–235.
- Ann Arbor Electrogram Libraries 2010. Ann Arbor MI, USA, <http://electrogram.com>.
- Barat, S., Kulkarni, V., Clark, T., & Barn, B. (2022). Digital twin as risk-free experimentation aid for techno-socio-economic systems. In *Proceedings of the 25th international conference on model driven engineering languages and systems* (pp. 66–75).

³ Parameter suggestion occurs one week after digital twin learning

- Chen, C., Hua, Z., Zhang, R., Liu, G., & Wen, W. (2020). Automated arrhythmia classification based on a combination network of CNN and LSTM. *Biomedical Signal Processing and Control*, 57, Article 101819.
- Cheriyian, J., & Maheshwari, S. (1989). Analysis of preflow push algorithms for maximum network flow. *SIAM Journal on Computing*, 18(6), 1057–1086.
- David, A., Larsen, K. G., Legay, A., Mikučionis, M., Poulsen, D. B., Vliet, J. v., et al. (2011). Statistical model checking for networks of priced timed automata. In *International conference on formal modeling and analysis of timed systems* (pp. 80–96). Springer.
- Gao, Q., Naumann, M., Jovanov, I., Lesi, V., Kamaravelu, K., Grill, W. M., et al. (2020). Model-based design of closed loop deep brain stimulation controller using reinforcement learning. In *2020 ACM/IEEE 11th international conference on cyber-physical systems* (pp. 108–118).
- Gold, M. R., Ahmad, S., Browne, K., Berg, K. C., Thackeray, L., & Berger, R. D. (2012). Prospective comparison of discrimination algorithms to prevent inappropriate ICD therapy: primary results of the Rhythm ID Going Head to Head Trial. *Heart Rhythm*, 9(3), 370–377.
- Graves, A., Fernández, S., Gomez, F., & Schmidhuber, J. (2006). Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd international conference on machine learning* (pp. 369–376).
- Jiang, Z., Abbas, H., Jang, K. J., Beccani, M., Liang, J., Dixit, S., et al. (2016). In-silico Pre-clinical trials for implantable cardioverter defibrillators. (pp. 169–172).
- Jiang, Z. H., Pajic, M., & Mangharam, R. (2012). Cyber-physical modeling of implantable cardiac medical devices. *Proceedings of the IEEE*, 100(1), 122–137.
- Kalyakulina, A. I., Yusipov, I. I., Moskalenko, V. A., Nikolskiy, A. V., Kozlov, A. A., Kosonogov, K. A., et al. (2018). LU electrocardiography database: a new open-access validation tool for delineation algorithms. *arXiv preprint arXiv:1809.03393*.
- Khan, W., Muntimadugu, E., Jaffe, M., & Domb, A. J. (2014). Implantable medical devices. *Focal Controlled Drug Delivery*, 33–59.
- Kim, S., Hori, T., & Watanabe, S. (2017). Joint CTC-attention based end-to-end speech recognition using multi-task learning. In *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)* (pp. 4835–4839). IEEE.
- Korte, T., Köditz, H., Niehaus, M., Paul, T., & Tebbenjohanns, J. (2004). High incidence of appropriate and inappropriate ICD therapies in children and adolescents with implantable cardioverter defibrillator. *Pacing and Clinical Electrophysiology*, 27(7), 924–932.
- Laguna, P., Mark, R. G., Goldberg, A., & Moody, G. B. (1997). A database for evaluation of algorithms for measurement of QT and other waveform intervals in the ECG. In *Computers in cardiology 1997* (pp. 673–676). IEEE.
- Lai, M., Yang, H., Gu, J., Chen, X., & Jiang, Z. (2022). Digital-twin-based online parameter personalization for implantable cardiac defibrillators. In *2022 44th annual international conference of the IEEE engineering in medicine & biology society (EMBC)* (pp. 3007–3010). IEEE.
- Lin, R., Stanley, M. D., Ghassemi, M. M., & Nemati, S. (2018). A deep deterministic policy gradient approach to medication dosing and surveillance in the ICU. In *2018 40th annual international conference of the IEEE engineering in medicine and biology society* (pp. 4927–4931).
- Mohammadi, A., Asadi, H., Mohamed, S., Nelson, K., & Nahavandi, S. (2017). OpenGA, a C++ genetic algorithm library. In *Systems, man, and cybernetics (SMC), 2017 IEEE international conference on* (pp. 2051–2056). IEEE.
- Moskalenko, V., Zolotykh, N., & Osipov, G. (2019). Deep learning for ECG segmentation. In *International conference on neuroinformatics* (pp. 246–254). Springer.
- Moss, A. J., Schuger, C., Beck, C. A., Brown, M. W., Cannom, D. S., Wilber, D., et al. (2012). Reduction in inappropriate therapy and mortality through ICD programming. *New England Journal of Medicine*, 367(24), 2275–2283. <http://dx.doi.org/10.1056/NEJMoa1211107>, arXiv:<https://doi.org/10.1056/NEJMoa1211107>, PMID: 23131066.
- Müller, M. (2007). Dynamic time warping. *Information Retrieval for Music and Motion*, 69–84.
- Olde Nordkamp, L. R., Brouwer, T. F., Barr, C., Theuns, D. A., Boersma, L. V., Johansen, J. B., et al. (2015). Inappropriate shocks in the subcutaneous ICD: Incidence, predictors and management. *International Journal of Cardiology*, 195, 126–133.
- Olshansky, B., Atteya, G., Cannom, D., Heidbuchel, H., Wilhelm, M., Lampert, R., et al. (2019). Competitive athletes with implantable cardioverter-defibrillators—How to program? Data from the implantable cardioverter-defibrillator sports registry. *Heart Rhythm*, 16(4), 581–587.
- Peretto, G., Durante, A., Limite, L. R., Cianflone, D., et al. (2014). Postoperative arrhythmias after cardiac surgery: incidence, risk factors, and therapeutic management. *Cardiology Research and Practice*, 2014.
- Rahul, J., Sora, M., & Sharma, L. D. (2021). A novel and lightweight P, QRS, and T peaks detector using adaptive thresholding and template waveform. *Computers in Biology and Medicine*, 132, Article 104307.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv*.
- Tappler, M., Aichernig, B. K., Larsen, K. G., & Lorber, F. (2019). *Time to learn – learning timed automata from tests* (pp. 216–235). Springer International Publishing, http://dx.doi.org/10.1007/978-3-030-29662-9_13.
- Verwer, S. (2010). Efficient identification of timed automata. <https://api.semanticscholar.org/CorpusID:1639150>.
- Wang, X., Ma, L., Li, H., Yin, Z., Luan, T., & Cheng, N. (2022). Digital twin-assisted efficient reinforcement learning for edge task scheduling. In *2022 IEEE 95th vehicular technology conference (VTC2022-Spring)* (pp. 1–5). IEEE.
- Wang, Y., Scheinman, M. M., Chien, W. W., Cohen, T. J., Lesh, M. D., & Griffin, J. C. (1991). Patients with supraventricular tachycardia presenting with aborted sudden death: incidence, mechanism and long-term follow-up. *Journal of the American College of Cardiology*, 18(7), 1711–1719.
- Wang, L., Zhang, W., He, X., & Zha, H. (2018). Supervised reinforcement learning with recurrent neural network for dynamic treatment recommendation. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2447–2456).
- Wilkoff, B. L., Kuhlkamp, V., Volosin, K., Ellenbogen, K., Waldecker, B., Kacet, S., et al. (2001). Critical analysis of dual-chamber implantable cardioverter-defibrillator arrhythmia detection: results and technical considerations. *Circulation*, 103(3), 381–386.
- Wilkoff, B. L., Williamson, B. D., Holloman, K. K., et al. (2008). Strategic programming of detection and therapy parameters in implantable cardioverter-defibrillators reduces shocks in primary prevention patients: Results from the PREPARE (primary prevention parameters evaluation) study. *Journal of the American College of Cardiology*, 52(7), 541–550.
- Wu, Y., Tang, R., Kang, E., & Jiang, Z. (2021). *Model-based clinical assist system for cardiac ablation* (pp. 110–120). New York, NY, USA: Association for Computing Machinery, <http://dx.doi.org/10.1145/3450267.3450539>.
- Yochum, M., Renaud, C., & Jacquir, S. (2016). Automatic detection of P, QRS and T patterns in 12 leads ECG signal based on CWT. *Biomedical Signal Processing and Control*, 25, 46–52.
- Zhao, X., Zhang, L., Ding, Z., Xia, L., Tang, J., & Yin, D. (2018). Recommendations with negative feedback via pairwise deep reinforcement learning. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 1040–1048).